

COD-PS Viet Nam

Update Year 2024 / Reference Year 2024



Objective of This Explanatory Technical Note

This explanatory technical note provides supplementary notes on the strengths and limitations of each Common Operational Dataset on Population Statistics (COD-PS) against the projections by the WPP 2022 Revision of the UN to enable informed humanitarian decision-making.

Structure of Supplementary Note

This supplementary note is organized into the following sections:

1. Metadata
2. Methodological Documentation
3. Intrinsic Population Growth Rates (ADM-0 & 1)
4. Population counts by age and sex (ADM-0)
5. Relative population size by age and sex (ADM-1)

1. Metadata

Item	Metadata
Country	Viet Nam
Source(s)	General Statistics Office Viet Nam
Source(s) Link(s)	https://vietnam.unfpa.org/sites/default/files/pub-pdf/final_population_projection_eng_to_upload.pdf
Population Data Type (Census enumeration, Direct estimate, or Population projection)	Population projection
Year of the Baseline Population	2019
Reference year of this COD-PS	2024
Publication year of this COD-PS	2024
ADM-1 Name	Tinh (province), Thanh pho truc thuoc trung uong (centrally-run city)
ADM-1 Number of Units	63 (58 provinces and 5 centrally-run cities)
ADM-1 Sex and Age Disaggregation	Sex and age disaggregated by 5-year age groups

Item	Metadata
ADM-1 Open-ended Age Group	80+
General Notes	The respondents of the 2019 census are Vietnamese citizens, including those working or studying for extended periods abroad within the permitted period of time, are reported as part of their households. This means that, despite living overseas, they are still counted as usual residents in Viet Nam.
Data Limitations	The projections, on which this COD-PS is based, may be overestimating population growth by a small margin because the Viet Nam NSO counts Vietnamese citizens living overseas for an extended period of time as the 2019 census respondents, and in turn, includes them in the projections.
COD-PS Demographic Data Quality Assessment	The COD-PS was assessed against ADM-0 estimates and projections published in the World Population Prospects (2022 Revision) by the United Nations Department of Economic and Social Affairs (UN-DESA).

2. Methodological Documentation

Item	Methodological Documentation
Methodology Used	Cohort component method was applied to the 2019 census results to construct the projections on which this COD-PS is based.
Baseline Population	2019 census
Post-enumeration survey (PES)	The post enumeration survey (PES) results showed that the net coverage error of the 2019 census was 0.2%.
Assessment and Adjustment of the Baseline Population	Results of the PES show the 2019 census to be relatively complete. The total population count of the 2019 census was found to be close to both the UN Population Division's mid-2019 population estimate for Viet Nam (WPP 2019 Revision) and the Viet Nam NSO projection for 2019 based on the 2009 census. The quality of age information from the 2019 census (age heaping, index of age preference, age-sex accuracy) was found to be in good standing. The 2019 census also found that 98.8% of under-5 children were registered at birth; particularly, 95.6% of children were registered during the first 6 months after birth. Because of the evident high quality of the 2019 census at multiple levels (national, urban and rural areas, 6 socio-economic regions, and 63 provinces and centrally-run cities), the NSO did not adjust the baseline population used for these projections.
Fertility (births)	Fertility assumptions were constructed primarily based on the future total fertility rate (TFR) trend assumptions outlined in the WPP 2010 Revision in accordance with the Bi-Logistic Fertility Decline Model. The 2019 TFR's of the whole country (2.09), urban areas (1.83), and rural areas (2.26) were taken as the starting point to extrapolate up to 2069. Under the medium fertility variant, it is assumed that by 2069 Viet Nam will reach a national TFR of 2.01 children per woman. TFR in subnational areas initially below 2.0 is projected to increase to or remain stable at 1.9. TFR in areas with initially higher fertility (TFR > 2.2) will decline to 2.1, and TFR in

Item	Methodological Documentation
	areas initially around replacement fertility (2.0-2.2) will decrease slightly to 2.01 by 2069.
Mortality (deaths)	Mortality assumptions were constructed based on the increasing average life expectancy at birth across the country, from 68.2 years (1999) to 72.8 years (2009) and 73.6 years (2019). This trend was in accordance with the UN predictions published in the WPP 2010 Revision. The projections upon which the COD-PS is based presume a slower rate of increase in areas where 2019 life expectancy was relatively high (70+ for males or 75+ for females).
International migration (net migration)	The projections, on which this COD-PS is based, assume zero net international migration because the respondents of the census are Vietnamese citizens. Vietnamese citizens who are working, studying for extended periods abroad within the permitted period of time are reported as part of their households. This means that, despite living overseas, they are still counted as usual residents in Viet Nam. The data on international migration from the 2019 census results suggests that the number of foreigners residing as nationals in Viet Nam is too small to affect the population size of the country.
Internal migration (migration within country)	Internal migration is expected to occur mainly at intra-provincial level. Inter-provincial migration is expected to occur at a lower rate for each five-year period of the total projection period. The NSO developed internal migration scenarios based on 5 groups of inter-provincial migration patterns, classified by the net migration of each province measured in the 2019 census.

3. Intrinsic Population Growth Rates (ADM-0 & 1)

At ADM-0, the implied postcensal population growth rates of this COD-PS (0.92% per year: female 0.85%, male 1.00%) are generally consistent with the implied growth rates of the World Population Prospects (WPP) 2022 Revision (2019: 0.89%, 2020: 0.92%, 2021: 0.77%, 2022: 0.70%, 2023: 0.66%, 2024: 0.63%).

At ADM-1, the National Statistical Office (NSO) predicts a relatively wide variation of postcensal population growth rates across the 63 ADM-1 units (58 provinces and 5 centrally administered cities), from high positive growth to negative growth. The NSO predicts the highest postcensal population growth rates for the country's economic and industrial centres with focused inflow of foreign investment: Binh Duong (4.36% per year: female 4.22%, male 4.49%), Bac Ninh (2.57% per year: female 2.45%, male 2.68%), Da Nang city (2.53% per year: female 2.44%, male 2.62%), Ho Chi Minh city (1.90% per year: female 1.80%, male 2.01%), and Ha Noi (1.83% per year: female 1.81%, male 1.86%). In contrast, the NSO predicts the lowest population growth rates for the provinces in the Mekong River Delta Region with least economic development and recurring annual flooding: Ca Mau (-0.30% per year: female -0.35%, male -0.25%), Dong Thap (-0.35% per year: female -0.34%, male -0.36%), Hau Giang (-0.38% per year: female -0.34%, male -0.43%), Soc Trang (-0.44% per year: female -0.45%, male -0.42%), and An Giang (-0.74% per year: female -0.69%, male -0.80%).

ADM-1	Last Census, Female	Last Census, Male	CODPS, Female	CODPS, Male	PGR(%), Female	PGR(%), Male	PGR(%), Both
An Giang	960,782	947,570	928,139	910,447	-0.69	-0.80	-0.74
Ba Ria - Vung Tau	572,085	576,228	589,562	596,946	0.60	0.71	0.66
Bac Giang	898,798	905,152	939,499	951,364	0.89	1.00	0.95
Bac Kan	153,869	160,036	153,602	162,614	-0.03	0.32	0.15
Bac Lieu	453,264	453,972	464,304	466,079	0.48	0.53	0.50
Bac Ninh	692,780	676,060	782,952	772,869	2.45	2.68	2.57
Ben Tre	657,971	630,492	653,054	632,958	-0.15	0.08	-0.03
Binh Dinh	754,831	732,087	747,165	732,368	-0.20	0.01	-0.10
Binh Duong	1,206,555	1,220,006	1,490,233	1,526,978	4.22	4.49	4.36
Binh Phuoc	493,206	501,473	508,671	519,403	0.62	0.70	0.66
Binh Thuan	612,424	618,384	623,553	635,822	0.36	0.56	0.46
Ca Mau	589,575	604,901	579,280	597,255	-0.35	-0.25	-0.30
Can Tho city	622,628	612,543	658,839	650,245	1.13	1.19	1.16
Cao Bang	264,721	265,620	265,931	273,687	0.09	0.60	0.34
Da Nang city	575,328	558,982	650,067	637,215	2.44	2.62	2.53
Dak Lak	926,744	942,578	953,830	975,688	0.58	0.69	0.64
Dak Nong	301,455	320,713	323,000	343,123	1.38	1.35	1.36
Dien Bien	295,420	303,436	316,108	326,954	1.35	1.49	1.42
Dong Nai	1,543,765	1,553,342	1,662,592	1,689,340	1.48	1.68	1.58

ADM-1	Last Census, Female	Last Census, Male	CODPS, Female	CODPS, Male	PGR(%), Female	PGR(%), Male	PGR(%), Both
Dong Thap	800,274	799,230	786,732	785,037	-0.34	-0.36	-0.35
Gia Lai	755,258	758,589	800,626	807,152	1.17	1.24	1.21
Ha Giang	422,908	431,771	443,905	457,179	0.97	1.14	1.05
Ha Nam	433,049	419,751	439,510	431,738	0.30	0.56	0.43
Ha Noi	4,061,744	3,991,919	4,445,812	4,380,120	1.81	1.86	1.83
Ha Tinh	648,157	640,709	650,508	651,895	0.07	0.35	0.21
Hai Duong	952,210	940,044	980,907	974,568	0.59	0.72	0.66
Hai Phong city	1,020,747	1,007,767	1,058,714	1,052,024	0.73	0.86	0.79
Hau Giang	366,811	366,206	360,669	358,477	-0.34	-0.43	-0.38
Ho Chi Minh city	4,611,840	4,381,242	5,045,250	4,843,629	1.80	2.01	1.90
Hoa Binh	427,208	426,923	438,467	442,778	0.52	0.73	0.62
Hung Yen	625,914	626,817	647,736	656,905	0.69	0.94	0.81
Khanh Hoa	618,594	612,513	633,540	630,305	0.48	0.57	0.52
Kien Giang	849,831	873,236	857,650	887,088	0.18	0.31	0.24
Kon Tum	268,819	271,619	288,796	292,508	1.43	1.48	1.46
Lai Chau	227,099	233,097	245,575	253,225	1.56	1.66	1.61
Lam Dong	643,832	653,074	667,766	679,249	0.73	0.79	0.76
Lang Son	382,245	399,410	388,678	411,673	0.33	0.60	0.46
Lao Cai	359,114	371,306	379,963	396,095	1.13	1.29	1.21
Long An	846,473	842,074	891,985	894,135	1.05	1.20	1.12

ADM-1	Last Census, Female	Last Census, Male	CODPS, Female	CODPS, Male	PGR(%), Female	PGR(%), Male	PGR(%), Both
Nam Dinh	908,358	872,035	899,466	870,964	-0.20	-0.02	-0.11
Nghe An	1,654,890	1,672,901	1,712,531	1,749,793	0.68	0.90	0.79
Ninh Binh	493,153	489,334	501,697	502,400	0.34	0.53	0.44
Ninh Thuan	294,441	296,026	297,276	301,639	0.19	0.38	0.29
Phu Tho	736,817	726,909	759,265	756,904	0.60	0.81	0.71
Phu Yen	433,886	439,078	432,182	439,455	-0.08	0.02	-0.03
Quang Binh	446,134	449,296	451,687	458,576	0.25	0.41	0.33
Quang Nam	760,226	735,586	761,034	748,202	0.02	0.34	0.18
Quang Ngai	619,783	611,914	609,435	608,424	-0.34	-0.11	-0.22
Quang Ninh	648,802	671,522	672,302	695,913	0.71	0.71	0.71
Quang Tri	318,882	313,493	314,444	312,849	-0.28	-0.04	-0.16
Soc Trang	601,731	597,922	588,251	585,404	-0.45	-0.42	-0.44
Son La	615,817	632,598	653,063	674,352	1.17	1.28	1.23
Tay Ninh	584,985	584,180	581,659	584,460	-0.11	0.01	-0.05
Thai Binh	955,039	905,408	962,652	922,891	0.16	0.38	0.27
Thai Nguyen	657,554	629,197	695,946	671,353	1.13	1.30	1.21
Thanh Hoa	1,824,127	1,816,001	1,856,213	1,868,813	0.35	0.57	0.46
Thua Thien Hue	570,132	558,488	578,107	566,862	0.28	0.30	0.29
Tien Giang	898,364	865,821	915,521	883,935	0.38	0.41	0.40
Tra	512,310	496,858	510,585	500,251	-0.07	0.14	0.04

ADM-1	Last Census, Female	Last Census, Male	CODPS, Female	CODPS, Male	PGR(%), Female	PGR(%), Male	PGR(%), Both
Vinh							
Tuyen Quang	389,665	395,146	396,764	406,824	0.36	0.58	0.47
Vinh Long	518,913	503,878	515,842	503,083	-0.12	-0.03	-0.07
Vinh Phuc	577,533	573,621	598,273	597,977	0.71	0.83	0.77
Yen Bai	408,053	412,977	427,168	435,905	0.92	1.08	1.00
Total	48,327,923	47,881,061	50,434,533	50,334,364	0.85	1.00	0.92

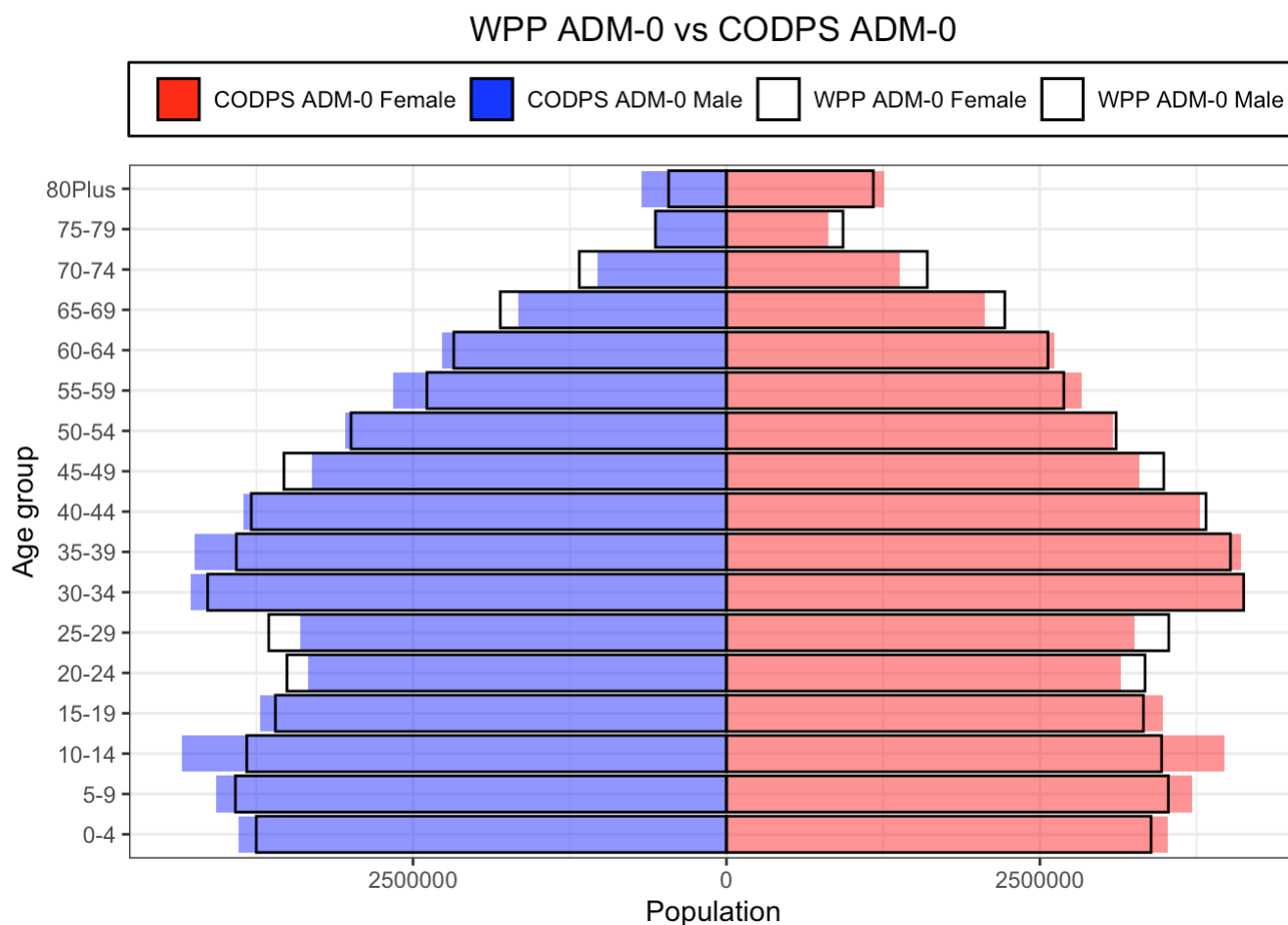
4. Population Counts by age and sex (ADM-0)

Below, we compare the population structure across sex and age groups between the NSO and WPP projections at the national level. At ADM-0, the age/sex structure of the population in this COD-PS is generally consistent with the WPP 2022 Revision projections.

The greatest relative differences in age- and sex-specific population counts (above 10%) include the 10-14 age group, where the NSO predicts higher numbers than the WPP (14.45% more females, 13.46% more males). The NSO also projects more males aged 55-59 (11.22%) and aged 80+ (46.6%). Compared to the WPP 2022 revisions, the NSO projects fewer persons ages 70-74 (-13.81% fewer females, -12.59% fewer males) and fewer females aged 75-79 (-12.53%).

	CODPS-WPP, Female	Difference(%)	CODPS-WPP, Male	Difference(%)
0-4	131,628	3.89	139,466	3.72
5-9	186,654	5.30	152,518	3.89
10-14	501,502	14.45	514,934	13.46
15-19	152,229	4.58	124,791	3.47
20-24	-189,974	-5.69	-168,928	-4.82
25-29	-275,988	-7.82	-251,080	-6.88
30-34	10,718	0.26	138,790	3.35
35-39	82,584	2.05	335,876	8.59
40-44	-45,287	-1.18	59,268	1.56
45-49	-197,225	-5.65	-222,243	-6.30
50-54	-22,476	-0.72	46,842	1.56
55-59	138,556	5.15	268,142	11.22
60-64	53,860	2.10	95,490	4.39

	CODPS-WPP, Female	Difference(%)	CODPS-WPP, Male	Difference(%)
65-69	-156,994	-7.07	-140,464	-7.79
70-74	-221,212	-13.81	-147,824	-12.59
75-79	-116,480	-12.53	-6,775	-1.19
80Plus	85,384	7.29	214,938	46.60



5. Relative population size by age and sex (ADM-1)

Below we compare the relative population proportions by sex and age between NSO ADM-1 and WPP ADM-0 projections. Reflecting the geographic and economic diversity of Viet Nam, the NSO predicts some variation in population structure across the 63 ADM-1 units with a few notable patterns emerging. First, the NSO predicts relatively a younger age structure with higher fertility and lower proportions of middle-aged and older adults compared to the WPP 2022 ADM-0 projection's for: Dak Nong, Dien Bien, Gia Lai, Ha Giang, Kon Tum, Lai Chau, Lao Cai, and Son La. Another observable pattern is among some ADM-1 units with low to negative growth rates, where the population is older and more heavily concentrated in ages 40+. These include: Ben Tre, Dong Thap, Tay Ninh, Thai Binh, Tien Giang, and Vinh Long. Additionally, The NSO also predicts distinctly higher proportions of young working adults than the WPP for the following ADM-1 units: Bac Ninh, Binh Duong, Dong Nai, Da Nang city, and Ho Chi Minh city.

